Elephant

Elephants are the largest living land animals. Three living species are currently recognised: the African bush elephant (*Loxodonta africana*), the African forest elephant (*L. cyclotis*), and the Asian elephant (*Elephas maximus*). They are the only surviving members of the family Elephantidae and the order Proboscidea; extinct relatives include mammoths and mastodons. Distinctive features of elephants include a long proboscis called a trunk, tusks, large ear flaps, pillar-like legs, and tough but sensitive grey skin. The trunk is prehensile, bringing food and water to the mouth and grasping objects. Tusks, which are derived from the incisor teeth, serve both as weapons and as tools for moving objects and digging. The large ear flaps assist in maintaining a constant body temperature as well as in communication. African elephants have larger ears and concave backs, whereas Asian elephants have smaller ears and convex or level backs.

Elephants are scattered throughout sub-Saharan Africa, South Asia, and Southeast Asia and are found in different habitats, including savannahs, forests, deserts, and marshes. They are herbivorous, and they stay near water when it is accessible. They are considered to be keystone species, due to their impact on their environments. Elephants have a fission–fusion society, in which multiple family groups come together to socialise. Females (cows) tend to live in family groups, which can consist of one female with her calves or several related females with offspring. The leader of a female group, usually the oldest cow, is known as the matriarch.

Males (bulls) leave their family groups when they reach puberty and may live alone or with other males. Adult bulls mostly interact with family groups when looking for a mate. They enter a state of increased testosterone and aggression known as musth, which helps them gain dominance over other males as well as reproductive success. Calves are the centre of attention in their family groups and rely on their mothers for as long as three years. Elephants can live up to 70 years in the wild. They communicate by touch, sight, smell, and sound; elephants use infrasound and seismic

Elephants
Temporal range:

A female African bush elephant in Mikumi National Park, Tanzania

Scientific classification	
Kingdom:	Animalia
Phylum:	Chordata
Class:	Mammalia
Infraclass:	Placentalia
Order:	Proboscidea
Superfamily:	Elephantoidea
Family:	Elephantidae

Living genera

- *Loxodonta* Anonymous, 1827
- *Elephas* Linnaeus, 1758

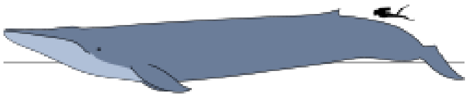
WIKIPEDIA
25 years of the free encyclopedia

Blue whale

Blue whale

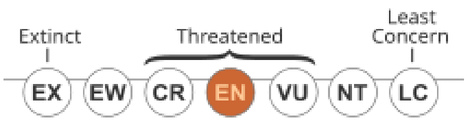
Temporal range: Early Pleistocene – Recent

Adult blue whale (*Balaenoptera musculus*)



Size compared to an average human

Conservation status



Endangered (IUCN 3.1)[2]

Vulnerable (NatureServe)[3]

CITES Appendix I[4]

Scientific classification

Kingdom:	Animalia
Phylum:	Chordata
Class:	Mammalia
Infraclass:	Placentalia
Order:	Artiodactyla
Infraorder:	Cetacea
Parvorder:	Mysticeti
Family:	Balaenopteridae
Genus:	*Balaenoptera*
Species:	***B. musculus***

The **blue whale** (***Balaenoptera musculus***) is a species of baleen whale and the largest marine mammal in the rorqual family Balaenopteridae. Reaching a maximum confirmed length of 29.9–30.5 m (98–100 ft) and weighing up to 190–200 t (190–200 long tons; 210–220 short tons), it is the largest animal known to have ever existed.[a] The blue whale's long and slender body can be of various shades of greyish-blue on its upper surface and somewhat lighter underneath. Four subspecies are recognized: *B. m. musculus* in the North Atlantic and North Pacific, *B. m. intermedia* in the Southern Ocean, *B. m. brevicauda* (the pygmy blue whale) in the Indian Ocean and South Pacific Ocean, and *B. m. indica* in the Northern Indian Ocean. There is a population in the waters off Chile that may constitute a fifth subspecies.

In general, blue whale populations migrate between their summer feeding areas near the poles and their winter breeding grounds near the tropics. There is also evidence of year-round residencies, and partial or age- and sex-based migration. Blue whales are filter feeders; their diet consists almost exclusively of krill. They are generally solitary or gather in small groups, and have no well-defined social structure other than mother–calf bonds. Blue whales vocalize, with a fundamental frequency ranging from 8 to 25 Hz; their vocalizations may vary by region, season, behavior, and time of day.

The blue whale was abundant in nearly all the Earth's oceans until the end of the 19th century. It was hunted almost to the point of extinction by whalers until the International Whaling Commission banned all blue whale hunting in 1966. The International Union for Conservation of Nature has listed blue whales as endangered as of 2018. Blue whales continue to face numerous man-made threats such as ship strikes, pollution, ocean noise, and climate change.

Taxonomy

Nomenclature

The genus name, *Balaenoptera*, means *winged whale*,[8] while the species name, *musculus*, could mean "muscle" or a diminutive form of "mouse", possibly a pun by Carl Linnaeus[8][9] when he named the species in *Systema Naturae*.[10] One of the first published descriptions of a blue whale comes from Robert Sibbald's *Phalainologia Nova*,[11] after Sibbald found a stranded whale in the estuary of the Firth of Forth, Scotland, in 1692. The name "blue whale" was derived from the

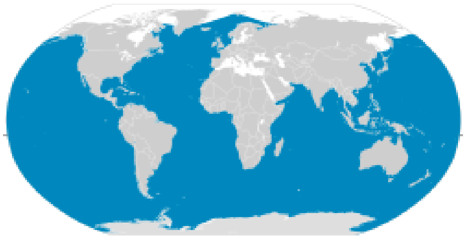
Binomial name

Balaenoptera musculus

(Linnaeus, 1758)

Subspecies

- *B. m. brevicauda* Ichihara, 1966
- *?B. m. indica* Blyth, 1859
- *B. m. intermedia* Burmeister, 1871
- *B. m. musculus* Linnaeus, 1758



Blue whale range (in blue)

Synonyms

- *Balaena musculus* Linnaeus, 1758
- *Balaenoptera gibbar* Scoresby 1820
- *Pterobalaena gigas* Van Beneden 1861
- *Physalus latirostris* Flower 1864
- *Sibbaldius borealis* Gray 1866
- *Flowerius gigas* Lilljeborg 1867
- *Sibbaldius sulfureus* Cope 1869
- *Balaenoptera sibbaldii* Sars 1875

Norwegian *blåhval*, coined by Svend Foyn shortly after he had perfected the harpoon gun. The Norwegian scientist G. O. Sars adopted it as the common name in 1874.[12]

Blue whales were referred to as "Sibbald's rorqual", after Robert Sibbald, who first described the species.[11] Whalers sometimes referred to them as "sulphur bottom" whales, as the bellies of some individuals are tinged with yellow.[13] This tinge is due to a coating of huge numbers of diatoms.[13] (Herman Melville briefly refers to "sulphur bottom" whales in his novel *Moby-Dick*.[14])

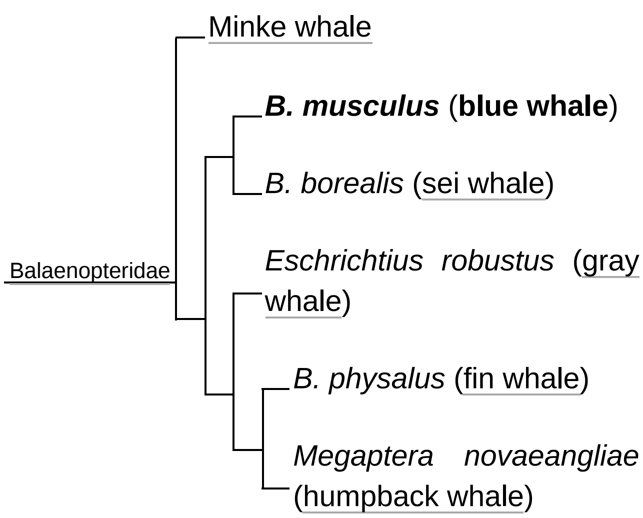
Evolution

Blue whales are rorquals in the family Balaenopteridae. A 2018 analysis estimates that the Balaenopteridae family diverged from other families in between 10.48 and 4.98 million years ago during the late Miocene.[15] The earliest discovered anatomically modern blue whale is a partial skull fossil from southern Italy identified as *B.* cf. *musculus*, dating to the Early Pleistocene, roughly 1.5–1.25 million years ago.[1] The Australian pygmy blue whale diverged during the Last Glacial Maximum. Their more recent divergence has resulted in the subspecies having a relatively low genetic diversity,[16] and New Zealand blue whales have an even lower genetic diversity.[17]

Whole genome sequencing suggests that blue whales are most closely related to sei whales with gray whales as a sister group. This study also found significant gene flow between minke whales and the ancestors of the blue and sei whale. Blue whales also displayed high genetic diversity.[15]

Hybridization

Blue whales are known to interbreed with fin whales.[18] The earliest description of a possible hybrid between a blue whale and a fin whale was a 20 m (66 ft) anomalous female whale with the features of both the blue and the fin whales taken in the North Pacific.[19] A whale captured off northwestern Spain in 1984, was found to have been the product of a blue whale mother and a fin whale father.[20]



Two live blue-fin whale hybrids have since been documented in the Gulf of St. Lawrence (Canada), and in the Azores (Portugal).[21] DNA tests done in Iceland on a blue whale killed in July 2018 by the Icelandic whaling company Hvalur hf., found that the whale was the offspring of a male fin whale and female blue whale;[22] however, the results are pending independent testing and verification of the samples. Because the International Whaling Commission classified blue whales as a "Protection Stock", trading their meat is illegal, and the kill is an infraction that must be reported.[23] Blue-fin hybrids have been detected from genetic analysis of whale meat samples taken from Japanese markets.[24] Blue-fin whale hybrids are capable of being fertile. Molecular tests on a 21 m (70 ft) pregnant female whale caught off Iceland in 1986 found that it had a blue whale mother and a fin whale father, while its fetus was sired by a blue whale.[25]

A phylogenetic tree of six baleen whale species[15]

In 2024, a genome analysis of North Atlantic blue whales found evidence that approximately 3.5% of the blue whales' genome was derived from hybridization with fin whales. Gene flow was found to be unidirectional from fin whales to blue whales. Comparison with Antarctic blue whales showed that this hybridization began after the separation of the northern and southern populations. Despite their smaller size, fin whales have similar cruising and sprinting speeds to blue whales, which would allow fin males to complete courtship chases with blue females.[26]

There is a reference to a humpback–blue whale hybrid in the South Pacific, attributed to marine biologist Michael Poole.[8][27][28]

Subspecies and stocks

At least four subspecies of blue whale are traditionally recognized, some of which are divided into population stocks or "management units".[29][30][31] Like many large rorquals, the blue whale is a cosmopolitan species.[32] They have a worldwide distribution, but are mostly absent from the Arctic Ocean and the Mediterranean, Okhotsk, and Bering Sea.[29]

Aerial view of adult blue whale

- Northern subspecies (*B. m. musculus*)
  - North Atlantic population – This population is mainly documented from New England along eastern Canada to Greenland, particularly in the Gulf of St. Lawrence, during summer, though some individuals may remain there all year. They also aggregate near Iceland and have increased their presence in the Norwegian Sea. They are reported to migrate south to the West Indies, the Azores and northwest Africa.[29]
  - Eastern North Pacific population – Whales in this region mostly feed off California's coast from summer to fall and then Oregon, Washington State, the Alaska Gyre and Aleutian Islands later in the fall. During winter and spring, blue whales migrate south to the waters of Mexico, mostly the Gulf of California, and the Costa Rica Dome, where they both feed and breed.[29]
  - Central/Western Pacific population – This stock is documented around the Kamchatka Peninsula during the summer; some individuals may remain there year-round. They have been recorded wintering in Hawaiian waters, though some can be found in the Gulf of Alaska during fall and early winter.[29]
- Northern Indian Ocean subspecies (*B. m. indica*) – This subspecies can be found year-round in the northwestern Indian Ocean, though some individuals have been recorded travelling to the Crozet Islands during between summer and fall.[29]
- Pygmy blue whale (*B. m. brevicauda*)
  - Madagascar population – This population migrates between the Seychelles and Amirante Islands in the north and the Crozet Islands and Prince Edward Islands in the south where they feed, passing through the Mozambique Channel.[29]
  - Australia/Indonesia population – Whales in this region appear to winter off Indonesia and migrate to their summer feeding grounds off the coast of Western Australia, with major concentrations at Perth Canyon and an area stretching from the Great Australian Bight and Bass Strait.[29]